ASCENTGEN BIBLE (MASTER)

A Ray-Peat-inspired instrument for generative ascent

1. Origin and problem

1.1 Personal origin story

Ascentgen began as a survival tool, not a startup concept.

I'm a man in my early 30s, a freelance video editor, who spent years trying to fix hair loss, weight gain, psoriasis, and collapsing energy with protocols, supplements, and "biohacking." My labs often looked "normal," but my life did not. I gained roughly 35 pounds, my hair loss and psoriasis accelerated, my CO_2 dropped, and I slid into a low-energy, low-resilience state that felt like accelerated aging.

Deeper testing eventually showed the underlying picture: functional hypothyroidism with a T4→rT3 conversion issue and steroidogenic collapse (very low pregnenolone, progesterone, and DHEA-S). The system was running on stress hormones, scraping by.

Discovering Ray Peat's work pulled this into a coherent frame. Instead of treating hair, skin, and weight as isolated "problems," the focus shifted to restoring systemic metabolic function: thyroid, oxidative metabolism, CO_2 production, and protective steroids, over a realistic time frame.

The simplest, most powerful practical change was tracking waking and post-meal temperature and pulse every day. Those four numbers became my most honest feedback loop:

- When they warmed and steadied, I felt warmer, calmer, clearer, more resilient.
- When they fell or swung, sleep, mood, digestion, and hair followed.

Ascentgen is built from that experience. It exists to give others the same clear signal: a way to see whether they are moving out of dormancy and suppression into a more generative, ascending state.

1.2 What Ascentgen is

Ascentgen is a minimalist metabolic dashboard focused on four readings:

- Waking temperature
- Post-meal temperature
- Waking pulse
- Post-meal pulse

It:

- Makes these readings easy to log every day.
- Shows clean trends over time (especially 14–30 days).
- Computes a Generative Quotient (GQ) score over recent days (0–100) that reflects how generative your metabolism appears.
- Describes your current pattern in plain language (Cold & Slow, Hot but Wired, Warm & Steady), mapped to three bands: Dormant, Kindling, and Ascending.

Ascentgen does not aim to be an all-in-one health OS. It is a focused instrument panel for heat, rhythm, and generative ascent.

In its first phase, Ascentgen assumes you are using a simple glass or digital thermometer and manual pulse counting. In later phases, Ascentgen may pair with a dedicated Ascentgen Probe that automates these readings and sends them into the app after each measurement. The philosophy is the same: minimal friction, maximal clarity.

2. Philosophy and scope

2.1 Core beliefs

Ascentgen is built on a Ray-Peat-inspired bioenergetic view of health:

- Energy production is primary

  Health is not merely the absence of disease; it is the presence of energy that can create, repair, and differentiate. Adequate thyroid function and oxidative metabolism, with high CO_2 production, are central.

- Energy and structure are interdependent

  Structure depends on the quality and abundance of energy, and energy depends on healthy structure. You cannot separate them.

- Temperature and pulse are central, practical markers

  Waking and post-meal temperature and pulse are the most accessible, day-to-day window into thyroid adequacy, oxidative metabolism, and the balance between generative energy and stress chemistry.

- Generative energy > heroic suffering

  Regeneration comes from sufficient, well-organized energy, not from chronic deprivation or heroic stress. Ascent is not a grind; it's the return of warmth and coherence.

- Structure is medicine

  Sleep, routine, sobriety, spiritual practice, and emotional work are structural interventions that lower stress chemistry and support generative energy, not optional "lifestyle hacks."

- Hair, skin, and body composition are outputs

  Hair loss, graying, psoriasis, and fat gain reflect deeper metabolic and hormonal states. They are important, but they are outputs, not primary targets.

- Small, reversible moves

  Dietary and thyroid changes are best made slowly, one lever at a time, and evaluated over 7–14 days with temp/pulse trends—not in response to single numbers.

- Tools must clarify, not confuse

  A health tool should reduce anxiety and noise, not add complexity, gamification, and fear.

2.2 What Ascentgen is not

Ascentgen is deliberately narrow:

- It is not a general "biohacking" dashboard.
- It is not a supplement store disguised as an app.
- It is not a diagnostic device or replacement for medical care.
- It is not an official Ray Peat product or endorsed by his estate.
- It is not a gamified "health score" toy.

Ascentgen is a calm, focused tracker for temp, pulse, and generative patterns, with minimal, principled interpretation.

2.3 One-sentence definition

Ascentgen helps you track waking and post-meal temperature and pulse, visualize your metabolic trends, and see your Generative Quotient (GQ) as you move through Dormant, Kindling, and Ascending.

2.4 Generative energy and ascent

In the Peat-inspired frame, health is not just the absence of disease, but the presence of energy that can create, repair, and differentiate—energy abundant enough to support wholeness rather than mere survival. That kind of energy depends on efficient oxidative metabolism: thyroid-supported respiration that produces plenty of carbon dioxide, protects against lactic, stress-driven states, and supports protective steroids like pregnenolone and progesterone.

In that frame, ascent is not hustle or self-optimization. It is the movement from cold, lactic, stress-dominated metabolism toward a warmer, CO_2-rich, generative state in which structure, meaning, and resilience can be restored. For an Orthodox Christian, it can also echo participation in the divine energies of God: leaving the numbness of spiritual and physiological torpor for a more living, responsive warmth.

Ascentgen takes this high-level idea and gives it a daily handle. Rising, stabilizing temperature and pulse are treated as small, practical signs that generative energy is increasing—that you are moving through Dormant, into Kindling, and eventually into Ascending.

3. Measurement protocol

Ascentgen only works if measurements are consistent and honest. This section defines how users should measure so the data is meaningful.

3.1 Temperature measurement

Goal: basal-style readings that reflect your resting state, not quick clinical checks.

Thermometer:

- Use a glass thermometer or a slower digital thermometer that can remain in place for several minutes.
- Avoid relying on instant forehead/ear readings for trends.

Choose one location and stick with it:

- Oral (under the tongue), or
- Axillary (underarm).

3.1.1 Waking temperature

- Take immediately upon waking, before getting out of bed or using your phone.
- If you must get up briefly, minimize activity and lie back down for a few minutes before measuring.

Oral:

- Place thermometer deep under the tongue.
- Close mouth fully.
- Keep it in place for several minutes (glass: ~10 minutes; digital: ignore quick beep and leave longer).
- Record the reading and “oral” as the location.

Axillary:

- Ensure armpit is dry.
- Place thermometer high in the armpit.
- Press arm firmly against torso.
- Keep it in place for 10–15 minutes (glass) or several minutes (digital).
- Record the reading and “axillary” as the location.

3.1.2 Post-meal temperature

- Measure 45–60 minutes after your first substantial meal of the day.
- That meal should include:
    - Meaningful protein
    - Carbohydrate
    - Some saturated fat
- Sit quietly for a few minutes before measuring.
- Use the same location and method as waking.

3.1.3 Reference directions for temperature

These are directions, not rigid targets:

- Waking temperature trending toward about 36.6°C / 97.8°F.
- Post-meal temperature trending toward about 37.0°C / 98.6°F.

Small daily fluctuations are normal. The trend over 7–14+ days is what matters.

3.2 Pulse measurement

Pulse must be measured consistently.

Where:

- Radial pulse (wrist), or
- Carotid pulse (side of neck, gently).

Use index and middle finger (never your thumb).

3.2.1 Waking pulse

- After waking temp, still lying down or sitting quietly.
- Find your pulse and count beats for a full 60 seconds.
- Breathe normally; don't consciously slow or speed your breathing.

3.2.2 Post-meal pulse

- Take at the same time as post-meal temperature (45–60 minutes after the first substantial meal).
- Sit quietly a few minutes before counting.
- Count beats for 60 seconds.

3.2.3 Reference directions for pulse

Directions, not strict targets:

- A calm resting pulse in the 60–85 bpm range, in a well-fed, warm person, often reflects adequate thyroid-driven output.
- Slightly higher pulse post-meal (e.g., +5–10 bpm) can be normal if it feels calm and non-anxious.

The key contrast is between:

- Calm, full, steady pulse vs.
- Thin, pounding, racing, or anxious pulse (stress-driven).

3.3 Consistency, notes, practice, and reminders

Consistency is the limiting factor.

Daily minimum:

- Waking temperature
- Waking pulse
- Post-meal temperature
- Post-meal pulse

Highly recommended notes:

- Sleep duration and quality

- Obvious stressors (travel, conflict, illness)
- Training (rest, light, heavy)
- Changes in diet, meds, thyroid, or supplements
- Key symptoms (digestion, mood, hair/skin, energy)

Ascentgen assumes you might only log these four numbers plus notes. Everything else is optional.

App reminders and alarms

Ascentgen can optionally remind you when to measure:

- Morning baseline reminder at your chosen wake window.
- Post-meal reminder 45–60 minutes after logging your first substantial meal.

Reminders are prompts, not commands. The more often you respond within the same window each day, the more meaningful your GQ trends become.

4. Patterns and interpretation

Ascentgen does descriptive, not diagnostic, interpretation. It gives clear patterns you can reflect on and, if needed, discuss with a practitioner.

4.1 Time horizons

Interpretation horizon:

- Main lens: rolling 14-day window for GQ and patterns.
- Weekly lens: 7-day summaries and week-to-week changes.
- Context: multi-month views later.

Individual days are noisy. Patterns emerge over weeks, not hours.

4.2 Three primary metabolic patterns

Ascentgen classifies your recent data into three broad patterns.

4.2.1 Cold & Slow – Dormant pattern

Characteristic signs:

- Waking temps often below ~36.4°C / 97.6°F.
- Post-meal temps only rise slightly or remain low.

- Waking and daytime pulse often in the 50s–60s, with sluggish energy and low mood.

Interpretation:

- Often indicates a Dormant band: suppressed metabolic state, under-fueled and/or functionally hypothyroid, with generative energy low or scattered.
- The system is producing less energy and CO_2 than it needs and may rely on stress hormones to cope.

Ascentgen label:

- Pattern: “Cold & Slow”.
- GQ band: typically Dormant.

4.2.2 Hot but Wired – Mis-fired energy pattern

Characteristic signs:

- Temperatures may look “normal” or even high at times.
- Pulse often 90+ bpm at rest, with anxious or pounding sensations.
- Subjective symptoms: anxiety, irritability, poor sleep, crashes after stress or intense training.

Interpretation:

- Often indicates mis-fired energy: stress chemistry (adrenaline, cortisol) driving heart rate and warmth more than calm thyroid support.
- The body appears “revved” on numbers but is not in a truly generative, coherent state.

Ascentgen label:

- Pattern: “Hot but Wired”.
- GQ band: can still be Dormant or early Kindling, depending on temps and stability.

4.2.3 Warm & Steady – Ascending pattern

Characteristic signs:

- Waking temps usually in the high 36s°C / high 97s°F.
- Post-meal temps regularly near 37.0°C / 98.6°F.
- Pulse mostly in the 70s–80s, calm and full but not jittery.
- Subjective: better sleep, digestion, mood, hair, and capacity for stress and training.

Interpretation:

- Indicates an Ascending band: warm, steady, generative state where oxidative metabolism and structure support each other.
- Energy isn't just present; it's organized in a way that supports repair, creativity, and resilience.

Ascentgen label:

- Pattern: "Warm & Steady".
- GQ band: typically Ascending.

5. Generative Quotient (GQ) – conceptual spec

5.1 Definition

GQ (Generative Quotient) is a 0–100 index that summarizes how generative your metabolism appears over a recent window (default: last 14 days), based on:

- Average waking temperature
- Average post-meal temperature
- Average waking pulse
- Average post-meal pulse
- Stability (how much they fluctuate)

GQ is:

- A compass, not a diagnosis.
- A way to see direction: whether you are moving through Dormant, Kindling, and into Ascending.
- Always interpreted alongside lived experience.

5.2 Target values

Ascentgen uses internal targets that approximate a warm, steady, generative state:

- Waking temp ideal: 36.6 °C (≈ 97.8°F).
- Post-meal temp ideal: 37.0 °C (≈ 98.6°F).
- Waking pulse ideal: 75 bpm.
- Post-meal pulse ideal: 82 bpm.

Temperatures in Fahrenheit are converted to Celsius internally.

5.3 Components (high-level)

Over the chosen window (e.g., 14 days), Ascentgen computes:

- A closeness score (0–1) for each metric vs its ideal (waking temp, post-meal temp, waking pulse, post-meal pulse).
- A stability score (0–1) based on standard deviations for those metrics (lower variability = higher stability).

It then combines:

- Temperature closeness (avg temps)
- Pulse closeness (avg pulses)
- Stability (overall steadiness)

into a 0–100 GQ, with weights favoring both “how close” and “how steady.”

5.4 GQ tiers and meanings

Tiers (conceptual):

- 0–69 → Dormant
- 70–89 → Kindling
- 90–100 → Ascending

Dormant (0–69)

Colder and/or unstable patterns. Heat and rhythm are not yet organized; generative energy is mostly dormant or scattered.

Kindling (70–89)

Temps and pulse are warming and becoming more coherent, but still somewhat fragile or variable. This is the “fire catching” band.

Ascending (90–100)

Temps and pulse are close to targets and stable. Reflects a warm, steady, generative pattern where energy production and structure support each other.

GQ is most useful as a story of movement across Dormant, Kindling, and Ascending over weeks and months, not as a single number to hit on a given day.

6. Product model and features (MVP)

6.1 Core data model

User

- id
- email / auth fields
- settings:
    - temperature_unit (“C” / “F”)
    - time_zone
    - notification preferences

DailyEntry

- id
- user_id
- date (YYYY-MM-DD)
- waking_temp_value (float)
- waking_temp_unit (“C” / “F”)
- waking_pulse_bpm (int)
- post_meal_temp_value (float, nullable)
- post_meal_temp_unit (“C” / “F”, nullable)
- post_meal_pulse_bpm (int, nullable)
- notes (text, optional)
- created_at, updated_at

GQSnapshot

- id
- user_id
- start_date (window start)
- end_date (window end)
- gq_score (int, 0–100)
- gq_tier (“Dormant”, “Kindling”, “Ascending”)
- created_at

6.2 Free features (non-negotiable core)

Free forever:

Logging

- Waking temp & pulse.
- Post-meal temp & pulse.
- Notes.

Visualization

- List/table of last 14–30 days of entries.
- Simple line charts (14–30 days) for:
    - Waking temperature
    - Post-meal temperature
    - Waking pulse
    - Post-meal pulse

GQ & patterns

- Rolling 14-day GQ calculation.
- GQ tier label: Dormant / Kindling / Ascending.
- Weekly pattern tag: Cold & Slow, Hot but Wired, Warm & Steady.
- Short explanations pulled from copy bank.

Weekly summary

- "This week vs last week" averages for temps and pulse.
- GQ change (e.g., 64 → 78) with a simple arrow and one-line commentary.

6.3 Future paid layers (concept only)

Paid layers, if developed, sit on top of the free core:

- Advanced views: 3-, 6-, 12-month charts; phase overlays.
- Reports & exports: PDFs, CSVs.
- Guided programs: e.g., 14-day baseline, 6–8 week GQ ascent cohorts.
- Contextual suggestion modules framed as options, not prescriptions.

Core tracking, trends, and GQ remain free.

6.4 Hardware: Ascentgen Probe (concept)

Ascentgen is designed to work with manual readings and to stand alone as a web app. In a future phase, it may integrate with a dedicated hardware device.

Ascentgen Probe (concept only):

- Oral thermometer + pulse sensor that captures temperature and pulse simultaneously under the tongue.
- Uses a high-precision thermistor for temperature and an optical PPG sensor for pulse, similar to medical pulse oximeter technology.
- Built-in vibration motor signals when a stable reading is reached (~60–120 seconds).
- Bluetooth Low Energy (BLE) is disabled while in-mouth; during measurement the probe behaves as a passive sensor with local storage only.

- After removal, the device briefly wakes the BLE radio, transmits the reading to Ascentgen in a short burst, and powers the radio back down.
- Rechargeable battery with USB-C charging, ideally via a small desk/nightstand dock matching the brand's gold accent.
- Device cannot be used while charging (hard lockout) for safety.

EMF stance

- Sensing first, transmission second.
- No active radio during in-mouth contact.
- Vibration is purely mechanical, not RF.

This hardware is a future, optional layer. The core app remains fully usable with simple thermometers and manual pulse counting.

7. Brand system

7.1 Visual identity

Feel

- Dark, contemplative, minimal.
- Feels like a quiet, serious instrument panel, not a playful fitness app.

Color palette (example hexes)

- Background: near-black with warm undertone (#050509).
- Card/surface: dark slate (#101017).
- Main text: soft off-white (#E5E7EB).
- Quote/highlight text: warm cream (#F5E9C8).
- Accent gold: (#D3A84D).
- Subtle gold: (#C9973C).
- Lines/borders: graphite (#272738).

Logo and mark

- Wordmark: "Ascentgen" in uppercase serif, slightly tracked out.
- Mark: AG monogram – a circle with an ascending arrow/thermometer in the center and an apex triangle – used as favicon/app icon and as the central "instrument" motif.
- In UI, the main GQ gauge visually echoes this mark: circular gauge + central vertical thermometer bar.

7.2 Typography

Two families:

- Serif (brand, quotes, large phrases) – e.g., Cormorant Garamond, Playfair Display, EB Garamond.
- Sans-serif (UI and body) – e.g., Inter, Manrope.

Rules:

- Serif for logo, big brand headlines, key quotes.
- Sans-serif for all labels, numbers, buttons, and paragraphs.
- Use all-caps sans with extra tracking for small labels (“GQ HISTORY · 14 DAYS”).
- Keep UI text slightly larger than typical for a calm, non-rushed feel.

7.3 Layout and interaction

- Single main column centered on larger screens; full-width with padding on mobile.
- Ample negative space around charts and numbers.
- One primary CTA per screen (e.g., “Begin your ascent”, “Record today’s ascent”).
- Animations: simple fades or slides; no flashy motion.

8. Voice, copy, and ethics

8.1 Tone and personality

Ascentgen’s voice is:

- Calm, precise, slightly poetic but always clear.
- Serious about physiology without being clinical or cold.
- Empathetic to people who feel “broken” by complex health issues and normal labs.
- Explicit about limits and uncertainty.

It avoids:

- Hype, memes, and “secret hack” language.
- Fear-based marketing or catastrophizing.
- Over-promising.

8.2 Core phrases and language

Reusable lines:

- “Begin your ascent.”
- “Track heat and rhythm at your core.”

- “From Dormant to Ascending.”
- “Your Generative Quotient (GQ) traces Dormant, Kindling, and Ascending.”
- “Structure is medicine.”
- “Systemic metabolic repair is primary; local or cosmetic tools are secondary.”
- “Ascentgen listens to four readings: waking and post-meal temperature and pulse.”

8.3 Microcopy and patterns

Buttons

- “Begin your ascent” (landing).
- “Enter Ascentgen” (open app).
- “Record today’s ascent” (log screen).
- “Lock in today’s readings” (save form).
- “Review your last ascent” (GQ history).
- “Decode your GQ” (learn more).

Empty states

- “No readings yet. Start with tomorrow’s waking temperature and pulse.”
- “Give us seven days of readings and we’ll show you your first GQ.”
- “We can’t trace an ascent without data. Start small, stay consistent.”

Errors

- “We couldn’t save that entry. Please check your connection and try again.”
- “Something went wrong loading your readings. Refresh or try again in a moment.”
- “GQ needs at least seven days of readings. Keep logging and check back soon.”

Pattern blurbs

- Cold & Slow – Dormant

  “Colder temps, flatter response to meals, and lower pulse suggest a Cold & Slow pattern—a Dormant band where generative energy hasn’t lit up yet.”

- Hot but Wired – mis-fired energy

  “Temps may look fine, but a high, erratic pulse and wired feeling often mean mis-fired energy—a stress-driven mix of Dormant and early Kindling, more alarm than ascent.”

- Warm & Steady – Ascending

  “Warm, steady temps and a calm pulse suggest a Warm & Steady pattern—the

Ascending band where generative energy is supporting repair and life.”

8.4 Ethics and lines

Commitments

- Core tracking (temp/pulse logging, basic charts, GQ, pattern labels) stays free.
- Ascentgen will not present itself as a diagnostic tool or replacement for medical care.
- Ascentgen will not sell miracle cures or promise specific outcomes.
- Any advanced education or coaching is clearly optional and not required to benefit from the tracker.

Disclaimers

- Ascentgen is not medical care; users should work with clinicians for diagnosis and treatment.
- Ascentgen is inspired by, but not affiliated with, Ray Peat or his estate.
- Ascentgen’s role is to make one crucial signal—heat and rhythm—visible, so you can see whether your ascent is real.

9. Generative Quotient (GQ) – App implementation (2026-04-26)

Note: This section documents the current live implementation (v0.2) in the app. It is allowed to differ slightly from the more general conceptual spec in section 5 while we iterate. When they diverge, this section is the source of truth for code behavior.

Definition

Generative Quotient (GQ) is a 0–100 score that summarizes how closely and how stably waking and post-meal temperature and pulse match the generative targets over recent days (14, 30, 90).

It is not “Growth Quotient”; all language in app and docs should use Generative Quotient (GQ).

Targets used in code

All scoring is done in Celsius internally (UI may show Fahrenheit):

- Waking temp: 36.6 °C.
- Post-meal temp: 37.0 °C.
- Waking pulse: 75 bpm.
- Post-meal pulse: 82 bpm.

Data source

Supabase table daily_entries with:

- date, waking_temp_value/_unit, waking_pulse_bpm, post_meal_temp_value/_unit, post_meal_pulse_bpm, user_id.

The app converts Fahrenheit to Celsius for scoring, but preserves original units in storage.

Windows

- 14-day GQ (current main view): uses last 14 days of entries (requires ≥ 7 days).
- 30-day GQ.
- 90-day GQ.

Closeness to targets (v0.2)

For each metric:

- Temperatures:
  - score 1.0 if within ±0.2 °C of ideal.
  - score 0.0 if ≥ 0.9 °C away.
  - linear between 0.2 and 0.9 °C.
- Pulse:
  - score 1.0 if within ±3 bpm of ideal.
  - score 0.0 if ≥ 20 bpm away.
  - linear between 3 and 20 bpm.

Stability (standard deviation)

- Temperature SD:
  - 1.0 if SD ≤ 0.15 °C.
  - 0.0 if SD ≥ 0.6 °C.
  - linear between 0.15 and 0.6 °C.
- Pulse SD:
  - 1.0 if SD ≤ 3 bpm.
  - 0.0 if SD ≥ 12 bpm.
  - linear between 3 and 12 bpm.

Composite scores

- TempScore = average of waking/post-meal temp closeness.
- PulseScore = average of waking/post-meal pulse closeness.
- StabilityScore = average of the four stability scores.

Final GQ

- GQ_raw = 100 × (0.35 × TempScore + 0.35 × PulseScore + 0.30 × StabilityScore)
- Clamp to 0–100 and round to nearest integer.

Tiers

- GQ < 70 → “Dormant”.
- 70–89 → “Kindling”.
- ≥ 90 → “Ascending”.

Patterns (7-day)

- Uses 7-day averages + pulse SD.
- “Cold & Slow”: temps below warm band.
- “Warm & Steady”: temps warm, pulse moderate, low variability.
- “Hot but Wired”: pulse high and/or unstable even if temps warm.

Current app status

- v0.2 GQ scoring is implemented in gq.ts.
- Daily entries are stored via a Today Check-In form.
- Recent entries list + inline editing exists; minor dev-tooling issues to resolve before fully exposing.

1. Brand Overview – Ascentgen

Name: Ascentgen

Core Metric: Generative Quotient (GQ)

Positioning: Precision bioenergetic instrument for daily metabolic tracking.

Logo

- **Core mark: Stylized "A/G" logo with:**
  - **Circle = instrument / gauge**
  - **Vertical arrow = thermometer / ruler**
  - **Apex triangle = ascent / higher order**

Logo refinements to apply

- **Uniform stroke weight across circle, arrow, and triangle.**
- **Arrow shaft to visually double as a thermometer/ruler:**
  - **Subtle calibration ticks along the vertical line.**

- **Ensure "G" opening gap is clean and intentional.**
- **Test at small sizes (16x16, 32x32) for clarity.**

Color System

- **Brand Gold: used for logo, primary GQ gauge, key accents.**
- **Background: very dark slate/charcoal (not pure black).**
- **Text: off-white / light gray.**
- **Numbers: consider monospaced font for "instrument" feel.**

Usage Notes

- **Gold reserved for primary state (current GQ, primary CTA).**
- **UI overall aesthetic: "dry, stiff, precise" – like lab instrumentation, not lifestyle.**

2. Generative Quotient (GQ) – System Definition

Purpose

- **Single composite metric reflecting metabolic / generative state.**
- **Built from daily temperature + pulse (+ future inputs as needed).**

Inputs (v1)

- **Morning basal oral temperature.**
- **Morning resting pulse.**
- **Future candidates: sleep, subjective energy, etc.**

Output

- **GQ: single score with defined range (e.g., 0–100).**
- **Color coding and qualitative ranges (e.g., "Suboptimal", "Adaptive", "Optimal").**

Daily Workflow (Ideal)

1. **Alarm/reminder triggers at set time.**
2. **User inserts Ascentgen Probe (future) or uses glass thermometer + manual pulse (MVP).**

3. **Reading taken for ~60–120 seconds.**
4. **Data logged automatically (future) or manually (MVP).**
5. **GQ computed and displayed on dashboard.**

UX Representation

- **Central circular gauge tied visually to logo "G".**
- **Tap gauge → expand to details:**
  - **Temp vs target**
  - **Pulse vs target**
  - **Simple recent trend line.**

3. Ascentgen Web App – MVP Spec

1. Core Objective

- **Enable daily logging of temp + pulse.**
- **Compute and display GQ.**
- **Establish user habit and prove value before hardware exists.**

2. Primary Screens

A. Dashboard (Home)

- **Center: GQ gauge (large, circular).**
- **Below: Today's temp + pulse.**
- **Primary CTA: "Log today's reading".**
- **Secondary: small trend sparkline for last 7–30 days.**

B. Log Reading Screen

- **Fields:**
  - **Temperature (°F/°C)**
  - **Pulse (bpm)**
  - **Time of reading (default = now)**
  - **Context tag (optional): Morning Baseline / Pre-Workout / Post-Meal etc.**
- **Design for thumb use on phone:**
  - **Big inputs**
  - **Minimal text**
  - **Large "Save" button at bottom center.**

C. Reminders / Alarm Settings

- **User-set reminder times (e.g., Morning Baseline @ 7:00 AM).**
- **Option to create multiple protocols:**
  - **Morning Baseline**
  - **Pre-Workout**
  - **Pre-Bed**
- **Each protocol:**
  - **Time**
  - **Days of week**
  - **Notification text.**

3. Tech Notes

- **Web-first, mobile-optimized (PWA-able).**
- **Secure (HTTPS).**
- **Local + cloud storage considerations for health data.**

4. Ascentgen Probe – Hardware Spec (v0.1)

1. Product Concept

A Bluetooth-enabled oral thermometer + pulse sensor that:

- **Captures temperature + pulse simultaneously.**
- **Vibrates when reading is complete.**
- **Stores reading during measurement, then transmits to Ascentgen app after removal.**
- **Minimizes EMF by disabling radio while in-mouth [web:594].**

2. Functional Requirements

Sensors

- **High-precision thermistor for oral temp.**
- **Optical PPG sensor for pulse (similar to pulse oximeter tech) [web:584].**

Operation Flow

1. **User starts measurement via app or device button.**
2. **Probe inserted under tongue.**
3. **Radio off during measurement; only sensors + timer active [web:594].**
4. **Device vibrates when temp & pulse stabilize (~60–120s) [web:596].**
5. **User removes probe.**
6. **Device wakes BLE, transmits reading(s) to app in a short burst.**
7. **Radio switches off again.**

EMF Strategy

- **BLE kept in deep sleep during measurement.**
- **Transmit only after probe is out of mouth.**
- **Vibration is mechanical, not RF-based.**

Power

- **Preferred: Rechargeable battery with USB-C charging.**
- **UX: drop-in charging dock that matches brand gold.**
- **Hard lockout: device cannot be used while charging (safety / medical compliance).**

(Option B: coin cell battery – low priority for now.)

Usage Modes

- **Primary: Oral.**
- **Future: optional armpit mode, but v1 focuses on oral for consistency and accuracy [web:598][web:602].**

3. Manufacturing & Cost – High-Level

> **These are directional, to guide strategy, not quotes.**

Development Phases (Typical)

1. **Product Requirement Spec (this doc)**
   - **Work with a medical device design consultancy to refine [web:615][web:619].**
   - **Est. cost: roughly $20k–$50k depending on firm and depth [web:614][web:618].**
2. **Design & Prototyping**
   - **Industrial design + electronics + firmware for BLE, power, and sensors [web:616].**
   - **Est. cost range often $50k–$150k for a first functional prototype batch [web:614].**

3. **Regulatory Path Assessment**
    - **Likely at least Class I medical device, possibly Class II depending on claims [web:618].**
    - **Budget: $100k–$500k+ including testing, documentation, submission.**
4. **Manufacturing Setup**
    - **Tooling, test jigs, quality system, etc.**
    - **Unit cost will depend heavily on volume.**

Unit Cost Considerations (Very Rough)

- **BOM (electronics, sensors, battery, enclosure).**
- **Assembly and QA (medical factory).**
- **Packaging + dock.**
- **Small runs (hundreds) are expensive per unit.**
- **Economies of scale start in the thousands.**

4. Next Actions (Hardware)

- **Shortlist 3–5 design firms that do small medical devices [web:613][web:616].**
- **Prepare a 1–2 page Product Requirements Summary (can export from this page).**
- **Begin with feasibility call; do not sign anything until MVP web app traction is proven.**

5. Launch & Marketing – Ascentgen

1. Web App v1 Launch

- **Goal: prove that daily GQ tracking is valuable using manual inputs.**
- **Channels:**
    - **Personal network**
    - **Ray Peat / bioenergetic adjacent communities**
    - **Social (TikTok / Twitter / etc.)**

Key Messages

- **"A precise daily instrument for your metabolism."**
- **"Track temp + pulse; get one clear Generative Quotient."**
- **"Built to match Ray Peat-style bioenergetic tracking."**

2. Hardware Teaser (Future)

- **Position hardware as:**
    - **"The Ascentgen Probe – automates your temp + pulse protocol."**
- **Don’t promise dates until feasibility and costs are clearer.**

3. Content Drafts

- **Why I built Ascentgen.**
- **What GQ is.**
- **Why temp + pulse matter (Peat style).**
- **How the web app works now.**
- **Future vision: full lab-in-your-pocket with Bluetooth probe.**

6. Open Questions

- **Exact GQ formula and scaling.**
- **How many protocols (morning only vs multi-point day).**
- **Exact hardware classification (Class I vs II).**
- **Potential non-US regulatory needs.**
- **Ideal initial manufacturing volume for v1 probe.**

1. Positioning & One‑liners

Ascentgen is a minimalist metabolic dashboard inspired by Ray Peat’s generative energy ideas. It tracks waking and post‑meal temperature and pulse, shows you simple trends, and gives you a Generative Quotient (GQ) so you can see whether you’re moving from Dormant through Kindling into Ascending.

Ascentgen helps you track heat and rhythm at your core—waking and post‑meal temp and pulse—so you can see if your metabolism is actually generating more life, not just more data.

Track temp and pulse. See your Generative Quotient.

From Dormant to Ascending.

---

2. Headlines

Begin your ascent.

Track heat and rhythm at your core.

From Dormant to Ascending.

A clear line on your metabolism.

A Ray-Peat-inspired instrument for generative ascent.

---

3. Short Explanations

Why temperature and pulse (long)

Most tools chase steps, badges, and vague "health scores." Ascentgen ignores the noise and goes straight to the core: heat and rhythm. Your waking and post-meal temperature and pulse show whether you're running on thyroid-driven oxidative metabolism or on raw stress chemistry. When these readings rise and steady, generative energy returns; when they fall or swing, the body tilts back into dormancy and defense.

Why temperature and pulse (short)

Your core story is simple: how warm you run, and how your heart beats at rest. Waking and post-meal temperature and pulse are enough to show if you're moving toward ascent or sinking into dormancy.

What GQ is (primary)

Your Generative Quotient (GQ) is a 0–100 index of the last 14 days of temp and pulse readings. It shows whether you're sitting in Dormant, Kindling, or Ascending, and how stable your pattern is. It's a compass, not a diagnosis.

What GQ is (short)

GQ tracks how your last 14 days of temperature and pulse move through Dormant, Kindling, and Ascending, so you can see if your metabolism is actually generating more life—not just more data.

---

4. Pattern Descriptions

Cold & Slow – Dormant (long)

Your recent temps and pulse match a suppressed pattern. Waking temps sit on the colder side, post-meal temps barely rise, and your pulse tends to be lower or flat. Many people in this pattern feel sluggish, cold, and less resilient to stress. This is the Dormant band:

generative energy is low or scattered, and the body leans heavily on stress chemistry just to get through the day.

Cold & Slow – Dormant (short)

Colder temps, flatter response to meals, and lower pulse suggest a Cold & Slow pattern—a Dormant band where generative energy hasn't lit up yet.

Hot but Wired – mis-fired energy (long)

Your temps may look "normal" or even high at times, but your pulse is often fast and jumpy, especially at rest. This pattern commonly reflects mis-fired energy: adrenaline and cortisol driving heart rate and warmth instead of calm thyroid support. People often feel anxious, wired, and prone to crashes after stress or intense training. It's a stress-driven mix of Dormant and early Kindling—more alarm than ascent.

Hot but Wired – mis-fired energy (short)

Temps can look fine, but a high, erratic pulse and wired feeling often mean mis-fired energy—a stress-driven blend of Dormant and early Kindling, more alarm than ascent.

Warm & Steady – Ascending (long)

Your waking temps sit in the warm range, your post-meal temps consistently climb toward 37.0°C / 98.6°F, and your pulse stays in a calm 70s–80s band. Subjectively, this often feels like good sleep, solid digestion, better hair and skin, and the ability to handle stress without crashing. This is the Ascending band: generative energy is present and organized, supporting repair, creativity, and life.

Warm & Steady – Ascending (short)

Warm, steady temps and a calm pulse suggest a Warm & Steady pattern—the Ascending band where generative energy is supporting repair and life.

---

5. Buttons & CTAs

Begin your ascent.

Enter Ascentgen.

Return to the signal.

Record today's ascent.

Lock in today's readings.

Review your last ascent.

Decode your GQ.

Tune Ascentgen.

Exit the ascent.

---

6. Empty States

No readings yet. Start with tomorrow's waking temperature and pulse.

Give us seven days of readings and we'll show you your first GQ.

We can't trace an ascent without data. Start small, stay consistent.

---

7. Errors & System Messages

We couldn't save that entry. Please check your connection and try again.

Something went wrong loading your readings. Refresh or try again in a moment.

GQ needs at least seven days of readings. Keep logging and check back soon.

We couldn't calculate a pattern for this period. Try to record both waking and post-meal readings most days.

---

8. Tone Rules (Do / Don't)

Do

- **Speak calmly and plainly, with occasional clean, poetic lines.**
- **Emphasize direction and trend over perfection.**

- **Acknowledge uncertainty and individual differences.**
- **Honor sleep, structure, food, and thyroid support as real interventions.**
- **Encourage small, reversible changes judged over weeks.**

Don’t

- **Use fear‑based phrasing or catastrophizing.**
- **Promise cures or specific outcomes.**
- **Use aggressive marketing hype or “secret hack” language.**
- **Talk down to users or hide how GQ is calculated.**
- **Treat GQ as a moral grade or a game.**

1. Measurement Protocol (User‑Facing)

Temperature – waking

- **Take immediately on waking, before getting out of bed or using your phone.**
- **Choose either oral (under tongue) or axillary (underarm) and stick with it.**
- **Oral: deep under tongue, mouth closed, several minutes.**
- **Axillary: dry armpit, thermometer high in pit, arm pressed firmly, several minutes.**
- **Log value and location.**

Temperature – post‑meal

- **Take 45–60 minutes after your first substantial meal (protein + carbs + some saturated fat).**
- **Sit quietly a few minutes before measuring.**
- **Use the same location and method as waking.**

Pulse – waking

- **After waking temp, still lying or sitting quietly.**
- **Use index/middle finger on wrist or side of neck.**
- **Count beats for a full 60 seconds.**
- **Log bpm.**

Pulse – post‑meal

- **Same timing as post‑meal temperature (45–60 minutes after the first real meal).**
- **Sit quietly a few minutes before counting.**
- **Count for 60 seconds.**

Reference directions

- **Waking temp trending toward ~36.6°C / 97.8°F.**

- **Post-meal temp trending toward ~37.0°C / 98.6°F.**
- **Calm resting pulse in ~60–85 bpm range, slightly higher after meals, driven by fuel and thyroid rather than anxiety.**

---

2. Internal Data Expectations

- **At most one DailyEntry per calendar day per user.**
- **Each DailyEntry can include:**
  - **Waking temp, post-meal temp.**
  - **Waking pulse, post-meal pulse.**
  - **Notes.**
- **Missing readings are allowed, but:**
  - **GQ requires at least 7 days with readings in the 14-day window.**
  - **Stability is more reliable with 10–14 days.**

All internal calculations convert temperatures to Celsius.

---

3. Pattern Logic (Internal)

High-level rule of thumb (can be tuned later):

Cold & Slow – Dormant

- **14-day average waking temp noticeably below target (e.g., < 36.4°C).**
- **14-day average post-meal temp not approaching 37.0°C.**
- **Waking pulse often < ~65 bpm or very flat.**

Hot but Wired – mis-fired energy

- **Temps near targets or intermittently high.**
- **Waking or post-meal pulse average elevated (e.g., > ~85–90 bpm).**
- **Pulse variability high (large standard deviation).**

Warm & Steady – Ascending

- **Waking temp average near 36.6°C.**
- **Post-meal temp average near 37.0°C.**
- **Pulse averages in 70–85 bpm range.**
- **Low variability in both temp and pulse.**

This logic is used to assign the weekly pattern labels (Cold & Slow, Hot but Wired, Warm & Steady).

---

4. GQ Calculation Spec

Targets

- **Waking temp target $T_{w,ideal}=36.6$ °C**
- **Post‑meal temp target $T_{p,ideal}=37.0$ °C**
- **Waking pulse target $P_{w,ideal}=75$ bpm**
- **Post‑meal pulse target $P_{p,ideal}=82$ bpm**

All temperatures converted to Celsius internally.

Averages over last 14 days

- **$\bar{T_w}$: average waking temp**
- **$\bar{T_p}$: average post‑meal temp**
- **$\bar{P_w}$: average waking pulse**
- **$\bar{P_p}$: average post‑meal pulse**

Require at least 7 days with readings in the window to compute GQ.

Temperature closeness scores (0–1)

For waking:

- **$\Delta T_w=|\bar{T_w}-T_{w,ideal}|$**

- **If $\Delta T_w\le 0.3$ °C → $S_{Tw}=1$**

- **If $0.3<\Delta T_w<1.2$ °C:**

  $S_{Tw}=1-\frac{\Delta T_w-0.3}{1.2-0.3}$

- **If $\Delta T_w\ge 1.2$ °C → $S_{Tw}=0$**

For post‑meal:

- **$\Delta T_p=|\bar{T_p}-T_{p,ideal}|$**
- **Same thresholds and formula for S_{Tp}.**

Pulse closeness scores (0–1)

For waking:

- $\Delta Pw=\lvert \bar{Pw}-Pw,ideal \rvert \Delta Pw=\lvert Pw-Pw,ideal \rvert$
- **If** $\Delta Pw\le 5\Delta Pw\le 5$ **bpm →** $SPw=1SPw=1$
- **If** $5<\Delta Pw<255<\Delta Pw<25$ **bpm:**

  $SPw=1-\Delta Pw-525-5SPw=1-25-5\Delta Pw-5$

- **If** $\Delta Pw\ge 25\Delta Pw\ge 25$ **bpm →** $SPw=0SPw=0$

For post-meal:

- $\Delta Pp=\lvert \bar{Pp}-Pp,ideal \rvert \Delta Pp=\lvert Pp-Pp,ideal \rvert$
- **Same thresholds and formula for** $SPpSPp$.

Stability scores (0–1)

Standard deviations over 14 days:

- $SDTw,SDTp,SDPw,SDPpSDTw,SDTp,SDPw,SDPp$

Temps:

- **If** $SDT\le 0.2SDT\le 0.2$ **°C → 1**
- **If** $0.2<SDT<0.70.2<SDT<0.7$ **°C:**

  $Sstab,T=1-SDT-0.20.7-0.2Sstab,T=1-0.7-0.2SDT-0.2$

- **If** $SDT\ge 0.7SDT\ge 0.7$ **°C → 0**

Pulse:

- **If** $SDP\le 4SDP\le 4$ **bpm → 1**
- **If** $4<SDP<154<SDP<15$ **bpm:**

  $Sstab,P=1-SDP-415-4Sstab,P=1-15-4SDP-4$

- **If** $SDP\ge 15SDP\ge 15$ **bpm → 0**

Overall stability:

$$S_{stab}=(S_{stab,Tw}+S_{stab,Tp}+S_{stab,Pw}+S_{stab,Pp})/4$$

Combine into GQ

- **Temperature closeness:**

  $$ST=(STw+STp)/2$$

- **Pulse closeness:**

  $$SP=(SPw+SPp)/2$$

Weights:

- **Temperatures: 30%**
- **Pulses: 30%**
- **Stability: 40%**

Raw GQ:

$$GQ_{raw}=100\times(0.30\times ST+0.30\times SP+0.40\times S_{stab})$$

Clamped & rounded:

$$GQ = \mathrm{round}(\ \max(0,\ \min(100,\ GQ_{raw})\ )\ )$$

Tiers

- **0–69 → Dormant**
- **70–89 → Kindling**
- **90–100 → Ascending**

---

5. Interpretation Rules (User-Facing)

Dormant

“Colder and/or unstable readings suggest the Dormant band—generative energy is low or scattered, and stress chemistry tends to dominate. This is where food, sleep, and gentle structure do the most good.”

Kindling

“You’re warming up and getting more stable. This is the Kindling band, where consistent routines and small, reversible changes help the fire take and hold.”

Ascending

“You’re spending most of your time in a warm & steady pattern. This is Ascending: heat and pulse supporting real regeneration. The job now is to maintain, not chase extremes.”

GQ is recalculated daily using the last 14 days of readings.

1. Product Vision

Ascentgen is a minimalist metabolic tracking tool that helps people log waking and post-meal temperature and pulse, see their trends, and understand their Generative Quotient (GQ) as they move through Dormant, Kindling, and Ascending.

Goals:

- **Make temp/pulse tracking simple and habitual.**
- **Make trends and GQ intuitive at a glance.**
- **Reduce anxiety and over-complication around health data.**
- **Honor Ray Peat’s generative energy ideas in a practical, non-dogmatic way.**

---

2. Core Data Model

User

- **id**
- **email**
- **password / auth fields (or OAuth)**
- **settings:**
    - **temperature_unit (“C” / “F”)**
    - **time_zone**
    - **notification preferences (future)**

DailyEntry

- **id**
- **user_id**
- **date (YYYY-MM-DD)**
- **waking_temp_value (float)**

- **waking_temp_unit (“C” / “F”)**
- **waking_pulse_bpm (int)**
- **post_meal_temp_value (float, nullable)**
- **post_meal_temp_unit (“C” / “F”, nullable)**
- **post_meal_pulse_bpm (int, nullable)**
- **notes (text, optional)**
- **created_at**
- **updated_at**

GQSnapshot

- **id**
- **user_id**
- **start_date (14‑day window start)**
- **end_date (14‑day window end)**
- **gq_score (int, 0–100)**
- **gq_tier (“Dormant”, “Kindling”, “Ascending”)**
- **created_at**

---

3. Free Features (MVP)

Free, non‑negotiable core:

Logging

- **Waking temp & pulse.**
- **Post‑meal temp & pulse.**
- **Notes for context.**

Visualization

- **List/table of last 14–30 days of entries.**
- **Simple 14–30 day line charts for:**
  - **Waking temperature.**
  - **Post‑meal temperature.**
  - **Waking pulse.**
  - **Post‑meal pulse.**

(Charts can be added after Ascension if needed; list view + GQ is the alpha minimum.)

GQ & Patterns

- **Daily GQ computation over rolling 14 days.**

- **GQ tier: Dormant / Kindling / Ascending.**
- **Short explanation of the current band.**
- **Weekly pattern tag:**
  - **Cold & Slow (Dormant).**
  - **Hot but Wired (mis‑fired energy).**
  - **Warm & Steady (Ascending).**

Weekly Summary

- **This week vs last week averages for:**
  - **Waking/post‑meal temp.**
  - **Waking/post‑meal pulse.**
- **GQ change (e.g., 62 → 77) with simple directional arrow and one‑line commentary.**

---

4. Future Paid Layers (Concept Only)

Anything paid lives on top of the free tracker:

Advanced views

- **3‑, 6‑, 12‑month charts.**
- **Phase overlays (e.g., “cut”, “maintenance”, “thyroid protocol”).**

Reports & exports

- **PDF reports with charts, GQ history, and notes.**
- **CSV exports of raw data.**

Guided programs

- **14‑day baseline logging program.**
- **6–8 week “Dormant to Ascending” group journeys.**

Contextual suggestion modules

- **Pattern‑specific levers to consider (food, training, stress, thyroid timing), framed as options, not prescriptions.**

Core logging, trends, and GQ remain free.

---

5. Screens & Flows (v1)

Route: / – Landing

Sections:

- **Hero:**
  - **Title: “Ascentgen”.**
  - **Subline: 1–2 sentences from Copy Bank about tracking temp/pulse and GQ.**
  - **Primary CTA: “Begin your ascent” → /app.**
- **“Why temperature & pulse”:**
  - **2–3 sentences from Copy Bank explaining heat and rhythm.**
- **“What Ascentgen alpha does”:**
  - **3 bullets:**
    - **Log waking and post-meal readings.**
    - **View last 14 days of data.**
    - **See GQ and current band (Dormant/Kindling/Ascending).**

Route: /app – Dashboard

Sections/modules:

1. **GQ card**
   - **Displays:**
     - **GQ number.**
     - **Band (Dormant / Kindling / Ascending).**
     - **One short line from Measurement Specs (interpretation rules).**
2. **Today’s card**
   - **If no entry:**
     - **Text: “No readings for today yet. On waking, take your temperature and pulse before anything else. After your first real meal, repeat.”**
     - **Button: “Record today’s ascent”.**
   - **If entry exists:**
     - **Show waking + post-meal temp/pulse.**
     - **Pattern tag: Cold & Slow / Hot but Wired / Warm & Steady.**
     - **Short pattern line from Copy Bank.**
3. **14-day list**
   - **Table/cards:**
     - **Date.**
     - **Waking temp & pulse.**
     - **Post-meal temp & pulse (or “—”).**
     - **Notes icon/preview.**
4. **Log entry form**
   - **Fields:**
     - **Waking temperature (value + unit).**
     - **Waking pulse.**
     - **Post-meal temperature.**

        - Post-meal pulse.
        - Notes.
    - Helper text:
        - “Ascentgen isn’t chasing perfect numbers. It’s tracing your direction.”
    - Button: “Lock in today’s readings”.

---

6. Brand Visuals

Colors (example)

- Background: `#050509` (near-black, warm).
- Card: `#101017`.
- Main text: `#E5E7EB`.
- Quote / highlight: `#F5E9C8`.
- Accent gold: `#D3A84D`.
- Subtle gold: `#C9973C`.
- Borders: `#272738`.

Typography

- Serif (brand/quotes/headlines): Cormorant Garamond / Playfair / EB Garamond.
- Sans (UI/body): Inter / Manrope.

Usage:

- Serif: logo, main marketing headlines, quotes.
- Sans: all UI text, labels, numbers, body copy.

Layout Rules

- Dark background, central column on desktop; padded full-width on mobile.
- One primary CTA per screen.
- Generous spacing; no crowded charts.
- Subtle transitions only (fades/slides), nothing flashy.

---

1. Core Positioning

The GQ App is a bioenergetic utility tool designed for precise daily measurement, timing, and trend analysis. It is an instrument for bioenergetic self-observation—not a generic health tracker or content platform.

---

2. Pricing Tiers

We utilize a freemium model. The Free tier establishes the habit and utility, while Premium and Pro provide the analytical depth required for serious protocol adherence and professional coaching.

Plan	Monthly	Annual (Best Value)	Target Audience
Free	$0	$0	Individuals starting the protocol.
Premium	$4.99/mo	$49.99/yr	Serious self-trackers needing deep analytics.
Pro	$9.99/mo	$99.99/yr	Bioenergetic coaches managing multiple clients.

---

3. Feature Comparison Table

Category	Feature	Free	Premium	Pro
Instruments	Manual logging (Temp/Pulse)	✓	✓	✓
	Pulse timer / Stopwatch	✓	✓	✓
	Meal/Rest countdowns	✓	✓	✓
Insights	14-day GQ	✓	✓	✓
	30/90-day GQ	—	✓	✓
	Pattern detection	—	✓	✓
	Custom thresholds	—	✓	✓
	Advanced charting	—	✓	✓

Workflow	Recent entries list	✓	✓	✓
	Data export	Basic	Full	Full
	Sharable PDF reports	—	✓	✓
Coaching	Multi-client dashboard	—	—	✓
	Coach review/annotations	—	—	✓
	Client review mode	—	—	✓

---

4. Tier Definitions

Free: The Instrument

Goal: Daily utility and habit formation.

- **Focused on the core loop: measure, time, log.**
- **Provides enough data to see short-term generative trends.**
- **Essential tools (timer, stopwatch) included to make it a standalone utility.**

Premium: The Self-Tracker

Goal: Deeper understanding and pattern discovery.

- **Unlocks the 30/90-day windows to see real generative progress.**
- **Pattern detection helps users identify what inputs (meals, timing) drive the best metrics.**
- **Sharable reports for personal records or sharing with a coach.**

Pro: The Coach Workspace

Goal: Professional oversight and efficiency.

- **Built for bioenergetic coaches to monitor multiple clients.**
- **Review mode allows for batch feedback, reducing the time to interpret client logs.**
- **Tools for annotation and reporting help coaches provide value faster.**

---

5. Future Roadmap (Hardware & Merch)

Phased rollout plan:

1. **Phase 1 (App): Stable instrument, analytics, and coach dashboard.**
2. **Phase 2 (Hardware): Branded glass thermometer and digital stopwatch/timer.**
3. **Phase 3 (Merch): Community identity items (hats, shirts) once brand trust is established.**

---

6. How to Read Values

Standard operating procedure for the protocol:

Temperature (Glass Thermometer)

- **Read the number where the fluid column ends.**
- **If it lands between lines, estimate to the nearest tenth.**
- **Enter in the app as a number (e.g., 97.8 or 36.6).**

Pulse

- **Count beats for the timed interval (60 seconds recommended).**
- **Enter as a whole number (e.g., 72).**

General Rule

- **Consistency is key. Measure at the same time, under the same conditions, every day.**
- **The utility of the GQ relies on the reliability of the input data.**

---

7. Implementation Status (2026-04-26)

- **v0.2 GQ scoring implemented in Next.js/Supabase.**
- **Data storage: `daily_entries` table in Supabase.**
- **Current Development Focus: Debugging Next.js build cache; re-enabling recent entries inline editing in the UI.**
- **Future: Pro dashboard and hardware integration design.**

AI Workflow Checklist (Quick Reference)

1. Sources of truth

- **Keep Ascendgen Bible (Master) – 2026-04-26 as the main spec.**
- **Update this doc whenever a spec/logic/copy change is final.**

---

2. AI roles

Claude 3.5 Sonnet – Build Partner

- **Treat Claude as:**
- **Use Claude for:**

Perplexity – Reviewer/Editor/Researcher

- **Use Perplexity for:**

---

3. Standard setup for Claude

- **In Claude Project, pin a system prompt that says:**

(Use the long prompt we wrote earlier.)

---

4. Daily workflow

Step 1 – Define task in Notion

- **Write today's task under `Ascendgen – Today`.**

Step 2 – Build with Claude

- **Open Claude 3.5 Sonnet.**
- **Remind it of its role (or rely on pinned prompt).**
- **Tell it today's task.**
- **Let it:**
- **Copy Claude's best output into Notion under `Working Drafts` or into the relevant spec.**

Step 3 – Review here (Perplexity)

- **Paste Claude's draft here with a clear instruction:**
- **Apply edits and capture final version.**

Step 4 – Sync back

- **Update Ascendgen Bible / spec in Notion with final version.**
- **Tell Claude next session:**

---

5. Which AI to start with?

- **Need to create/modify code, flows, UI, or many copy options?**
- **Need to verify safety, coherence, Peat consistency, or finalize spec/copy?**

0. Core principles

- **Single source of truth:**

  The document "Ascentgen Bible (Master) – 2026‑04‑26" is the authoritative spec for philosophy, measurement, GQ, brand, and ethics.

- **Two AI roles:**

  - **Claude 3.5 Sonnet → build partner (bioenergetic‑informed product + dev).**
  - **Perplexity (here) → reviewer, editor, and research brain with citations.**
- **You are the integrator:**

  All final changes must be reflected in Notion so both AIs stay aligned.

---

1. Roles of each assistant

1.1 Claude 3.5 Sonnet – "Bioenergetic Product + Dev Partner"

Claude's role:

- **Expert bioenergetic assistant grounded in Ray‑Peat‑style physiology:**
  - **Temp/pulse as primary indicators of thyroid and oxidative metabolism.**

    - **CO_2, steroidogenesis, and stress chemistry context.**
- **Senior product + UX designer:**
    - **Minimalist, instrument‑like dashboards.**
    - **Flows that respect your protocol and measurement rules.**
- **Senior full‑stack engineer:**
    - **Next.js + TypeScript + Supabase + Tailwind.**
    - **Clean architecture, clear components, migration scripts, etc.**
- **Future hardware integration consultant:**
    - **Ascentgen Probe concept: oral temp + pulse sensor, BLE off in‑mouth, vibration on completion, USB‑C dock.**

Use Claude when you need:

- **New code, refactors, or file structures.**
- **UI layout, flows, and detailed component trees.**
- **Bulk copy exploration (multiple variants).**

1.2 Perplexity (GPT‑powered) – “Reviewer, Editor, Researcher”

My role:

- **Spec & copy editor: tighten, clarify, and organize your docs (like the Ascentgen Bible).**
- **Physiology + safety checker: verify logic around temp/pulse, thermometers, accuracy, EMF, etc., with citations.**
- **Consistency enforcer: make sure changes stay aligned with Ray‑Peat‑style principles and your 26‑week protocol.**

Use me when you need:

- **Confidence that something is physiologically coherent and ethically framed.**
- **A “second pass” on anything Claude produced.**
- **A clean final version to paste back into Notion.**

---

2. Standard prompt for Claude 3.5 Sonnet

Create a pinned note inside Claude’s Ascentgen Project with this prompt:

`You are my long‑term expert assistant for building Ascentgen.

ROLE

- **You are simultaneously:**
    - **A Ray‑Peat‑informed bioenergetic consultant (temperature/pulse, thyroid, CO_2, steroidogenesis).**
    - **A minimalist product + UX designer for a serious health instrument (not a gamified app).**
    - **A senior full‑stack engineer (Next.js + TypeScript + Supabase + Tailwind).**
    - **A future hardware integration advisor for a Bluetooth oral temp + pulse probe (BLE off in‑mouth, vibration on completion, USB‑C dock).**

PROJECT

- **Ascentgen is a web app that tracks:**
    - **Waking temperature**
    - **Post‑meal temperature**
    - **Waking pulse**
    - **Post‑meal pulse**
- **It computes a Generative Quotient (GQ, 0–100) based on closeness + stability to ideal targets over 14/30/90 days.**
- **It classifies patterns as:**
    - **Cold & Slow (Dormant)**
    - **Hot but Wired**
    - **Warm & Steady (Ascending)**
- **The app is web‑first, mobile‑optimized, and should feel like a precise laboratory instrument panel.**
- **Future hardware ("Ascentgen Probe") may automate readings with strict EMF discipline (radio off while in mouth).**

SOURCE OF TRUTH

- **Assume I have an "Ascentgen Bible (Master)" spec dated 2026‑04‑26.**
- **That document defines:**
    - **Philosophy, measurement protocol, and patterns.**
    - **GQ conceptual spec and current implementation.**
    - **Brand system, copy tone, and ethics.**
- **Do NOT contradict that spec. If something seems inconsistent, surface it and propose options instead of silently changing it.**

HOW TO WORK

- **Treat all temp/pulse and GQ outputs as descriptive and supportive, not diagnostic or curative.**
- **For product/UX:**
    1. **Give clear flows, component trees, and ready‑to‑paste copy.**
- **For engineering:**
    1. **Use modern Next.js + TS + Supabase patterns.**

    2. **Prefer clarity and maintainability over clever tricks.**
- **Start each task by:**
    1. **Restating what you think I'm asking for.**
    2. **Asking any critical clarifying questions.**
    3. **Proposing a short plan.**
    4. **Then producing detailed code/UX.**

INTEGRATION WITH ANOTHER AI

- **I also use a second AI (Perplexity, GPT‑based) as:**
    - **A reviewer/editor for specs, copy, and formulas.**
    - **A research checker for physiology, EMF, Ray‑Peat alignment.**
- **When I say I'll "take this to the other AI," respond by:**
    - **Summarizing your proposal cleanly.**
    - **Highlighting assumptions/open questions that should be checked.**

WAIT for me to give you today's specific task (e.g., dashboard layout, logging form, GQ refactor) before diving into solutions.`

---

3. Day‑to‑day workflow

3.1 Step 1 – Define the task in Notion

In your Notion workspace, create a simple "Today's Task" section under Ascentgen, e.g.:

- **"Finalize dashboard layout and copy."**
- **"Implement daily logging form and Supabase schema."**
- **"Refine GQ thresholds for v0.3."**

This keeps you anchored.

3.2 Step 2 – Build with Claude

1. **Open Claude 3.5 Sonnet in the Ascentgen Project.**
2. **Paste or reference the standard prompt (if not already pinned).**
3. **Tell it the specific task:**
    - **"Today's task: design the dashboard UI consistent with the Ascentgen Bible."**
4. **Let Claude:**
    - **Restate task → ask questions → propose plan → produce code/flows/copy.**

When you’re satisfied with a draft:

- **Copy the final output into Notion under a page like `Ascentgen — Working Drafts` or directly into the relevant spec section.**

3.3 Step 3 – Review/refine here (Perplexity)

When you have a draft from Claude that feels “close,” come here and:

1. **Paste the relevant section (code, logic, or copy).**
2. **Add a clear directive, for example:**
    - **“Please review this GQ logic for math/physiology consistency with Peat-style temp/pulse.”**
    - **“Please tighten this dashboard copy and ensure it matches Ascentgen tone.”**

I will:

- **Check the reasoning against physiological/clinical sources.**
- **Flag any EMF, thermometer, or wording issues (e.g., over-claiming, diagnostic language).**
- **Propose a cleaner, final version.**

3.4 Step 4 – Sync back into Notion and Claude

After we refine something here:

1. **Update the corresponding section of your Ascentgen Bible (Master) or dev spec in Notion.**
2. **Next time you use Claude, paste the updated version and say:**
    - **“Update your internal context: this is now the authoritative version of [GQ implementation / dashboard copy / measurement protocol]. Use this from now on.”**

That prevents divergence between tools.

---

4. Quick decision guide: which AI to use?

- **Start with Claude 3.5 Sonnet when:**
    - **You need to create or change code, flows, or UI.**
    - **You want lots of options and iteration.**

- **Start with Perplexity when:**
  - **You're unsure if an idea is physiologically sound or Peat‑consistent.**
  - **You want a final, polished spec or doc to paste into Notion.**
  - **You need citations or safety sanity checks.**

Whenever in doubt:

- **Rough‑draft with Claude → finalize and verify here → store in Notion.**

1. Tech stack

- **Frontend: Next.js (App Router) + TypeScript**
- **Styling: Tailwind CSS**
- **Backend: Supabase (Postgres + Auth + Row‑level security)**
- **Deployment: Vercel (or similar)**
- **Platform: Web‑first, mobile‑optimized (PWA‑capable later)**

---

2. High‑level features (MVP)

- **User auth (email/password or magic link)**
- **Today Check‑In form (waking + post‑meal temp/pulse + notes)**
- **Daily entries list (last 14–30 days)**
- **GQ calculation (14/30/90 day) and tier display**
- **Pattern label (Cold & Slow / Hot but Wired / Warm & Steady)**
- **Simple charts for the four metrics**
- **Basic reminder settings (time, days) – even if only stored, not yet fully scheduled**

---

3. Routes / pages (Next.js)

`/` – Landing

- **Hero: explanation of Ascentgen, GQ, and patterns.**
- **CTA: "Enter Ascentgen" (to `/app` or `/login`).**
- **Very minimal, instrument‑like.**

`/login` / `/signup`

- **Auth forms using Supabase Auth.**
- **Magic link or email/password.**

`/app` – Main dashboard (protected)

Sections:

1. Header
   - Brand mark, GQ tier label, date.
2. GQ Card
   - Current 14-day GQ score + tier (Dormant / Kindling / Ascending).
   - Pattern label (Cold & Slow / Hot but Wired / Warm & Steady).
   - One-line explanation.
3. Today Check-In
   - Form:
     - Waking temp (value + unit).
     - Waking pulse.
     - Post-meal temp (value + unit).
     - Post-meal pulse.
     - Notes (optional).
   - Button: "Lock in today's readings".
4. Recent Entries
   - Table/list for last 14 days:
     - Date, waking temp/pulse, post-meal temp/pulse.
   - Optional: inline edit.
5. Trends
   - Small charts (14–30 days) for each metric.
   - GQ history mini-chart.

`/settings` – Settings (protected)

- Units (°C / °F).
- Time zone.
- Reminder times:
  - Morning baseline window.
  - Post-meal reminder delay (45–60 min).

---

4. Database schema (Supabase)

Table: `profiles`

- `id` (uuid, PK, references auth.users.id)
- `email` (text)
- `created_at` (timestamp)
- `timezone` (text)

- `temperature_unit` (text, 'C' | 'F', default 'F')
- `morning_reminder_time` (time, nullable)
- `post_meal_reminder_offset_minutes` (int, default 60)

Table: `daily_entries`

- `id` (uuid, PK)
- `user_id` (uuid, FK → profiles.id)
- `date` (date)
- `waking_temp_value` (float, nullable)
- `waking_temp_unit` (text, 'C' | 'F' | null)
- `waking_pulse_bpm` (int, nullable)
- `post_meal_temp_value` (float, nullable)
- `post_meal_temp_unit` (text, 'C' | 'F' | null)
- `post_meal_pulse_bpm` (int, nullable)
- `notes` (text, nullable)
- `created_at` (timestamp, default now())
- `updated_at` (timestamp, default now())

Indexes

- Unique per user per date: (`user_id`, `date`).
- Index on `user_id` for fast queries.

Table: `gq_snapshots` (optional v1, nice for caching)

- `id` (uuid, PK)
- `user_id` (uuid, FK)
- `window_days` (int – 14, 30, 90)
- `start_date` (date)
- `end_date` (date)
- `gq_score` (int)
- `gq_tier` (text: 'Dormant' | 'Kindling' | 'Ascending')
- `pattern_label` (text: 'Cold & Slow' | 'Hot but Wired' | 'Warm & Steady')
- `created_at` (timestamp)

For v1, you can compute GQ on the fly and skip this table; add it later if performance becomes an issue.

---

5. GQ calculation (implementation notes)

- **Logic lives in `app/lib/gq.ts`.**
- **Uses 14/30/90-day windows of `daily_entries` for:**
  - **Waking/post-meal temp and pulse (converted to °C internally).**
  - **Closeness and stability scores (v0.2 thresholds from Bible).**
- **Returns:**
  - **`gqScore`, `gqTier`, `patternLabel`, and per-metric stats.**

When refining, always update:

1. **Ascentgen Bible – GQ implementation section.**
2. **Code in `gq.ts`.**

Then re-sync with Claude and here.

---

6. Frontend component map

Inside `/app` route:

- `components/GQCard.tsx`
  - **Displays GQ score, tier, pattern, and a one-liner.**
- `components/TodayCheckInForm.tsx`
  - **Fields for all four readings + notes.**
  - **Handles submission to `/api/log-today`.**
- `components/RecentEntriesList.tsx`
  - **Table/list with last 14 days.**
  - **Optional inline editing.**
- `components/MetricChart.tsx`
  - **Generic chart component for temp/pulse (pass metric + data).**
- `components/Layout/AppShell.tsx`
  - **Header, main column layout, basic navigation.**

---

7. API routes (Next.js)

`POST /api/log-today`

- **Auth required.**

- Upsert by (`user_id`, `date`).
- Body: waking/post-meal temps + units + pulses + notes.
- Returns updated `daily_entries` row.

`GET /api/daily-entries`

- Query params: `windowDays` (e.g., 14, 30).
- Returns recent entries ordered by `date DESC`.

`GET /api/gq`

- Query params: `windowDays`.
- Server computes GQ from `daily_entries` using `gq.ts`.
- Returns `score`, `tier`, `pattern`, and optionally the underlying stats.

(You can also compute GQ directly in server components instead of a dedicated API route; this is just a clean separation.)

---

8. Integration with AI workflow

Add this small subsection to remind yourself how to use the two AIs against this dev spec:

- When designing or changing routes, components, or DB schema:
    - Take this `Dev / Engineering` page + Ascentgen Bible to Claude 3.5 Sonnet.
- Ask Claude to:
    - Propose file organization and code.
    - Generate TS interfaces, Supabase migration SQL, and component skeletons.
- When Claude proposes anything that touches:
    - Measurement protocol.
    - GQ math.
    - Any health phrasing.

→ **Bring it here for review and final wording with citations.**

- **After review, update:**
  - **This page (Dev spec).**
  - **Ascentgen Bible (if conceptual change).**